CODE ALPHA

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INTERNSHIP REPORT

On

BIOINFORMATICS

Task 1: DNA/PROTEIN SEQUENCE ANALYSIS

Submitted by:

Faiqa Riaz

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DNA/PROTEIN SEQUENCE ANALYSIS

BLAST-Based Homology Study of Human Insulin

CodeAlpha Bioinformatics Internship — Task1

1. Introduction

Insulin is a peptide hormone produced by the beta cells of the pancreas, and it plays a central role in regulating blood glucose levels by promoting the uptake of glucose into muscle, liver, and adipose tissue. Because of its well-characterized structure and its critical physiological function, insulin is one of the most extensively studied proteins in molecular biology and serves as an excellent case study for sequence homology analysis.

This report presents a sequence-level investigation of the human insulin protein (isoform 2 precursor, NCBI Accession: NP_001035835.1). The primary objective is to retrieve the reference amino acid sequence from the NCBI Protein database and perform a Basic Local Alignment Search Tool (BLAST) analysis to identify homologous sequences across other species. The results are then used to examine the degree of evolutionary conservation of insulin across the animal kingdom.

2. Methodology

The analysis was carried out using the following steps:

- The NCBI Protein database (ncbi.nlm.nih.gov) was searched using the query "Homo sapiens insulin", and the reference sequence "insulin, isoform 2 precursor [Homo sapiens]" (Accession: NP_001035835.1, 200 amino acids) was selected.
- The amino acid sequence was retrieved in FASTA format directly from the NCBI record page.
- The FASTA sequence was submitted to the Protein BLAST (blastp) tool at blast.ncbi.nlm.nih.gov, searched against the ClusteredNR database using default parameters.
- The resulting alignments were reviewed in the Clusters view, and key metrics — Percent Identity, E-value, Query Coverage, and Alignment Score — were recorded for the top-scoring hits across a range of species.

BLAST works by locating short, high-scoring matching segments between the query sequence and sequences in the database, then extending these matches into full local alignments. Two statistics are central to interpreting the output: Percent Identity, which measures how similar two aligned sequences are, and the E-value, which estimates the likelihood that a match of this quality occurred by random chance — the closer the E-value is to zero, the more statistically significant the match.

3. Results

The BLAST search returned a large number of significant hits (E-values ranging from 2e-140 to 7e-21), spanning primates, other placental mammals, birds, reptiles, and even a distantly related bacterial protein. Figures 1 and 2 show the BLAST job summary and the ranked results table respectively. Table 1 summarizes representative hits across this range.

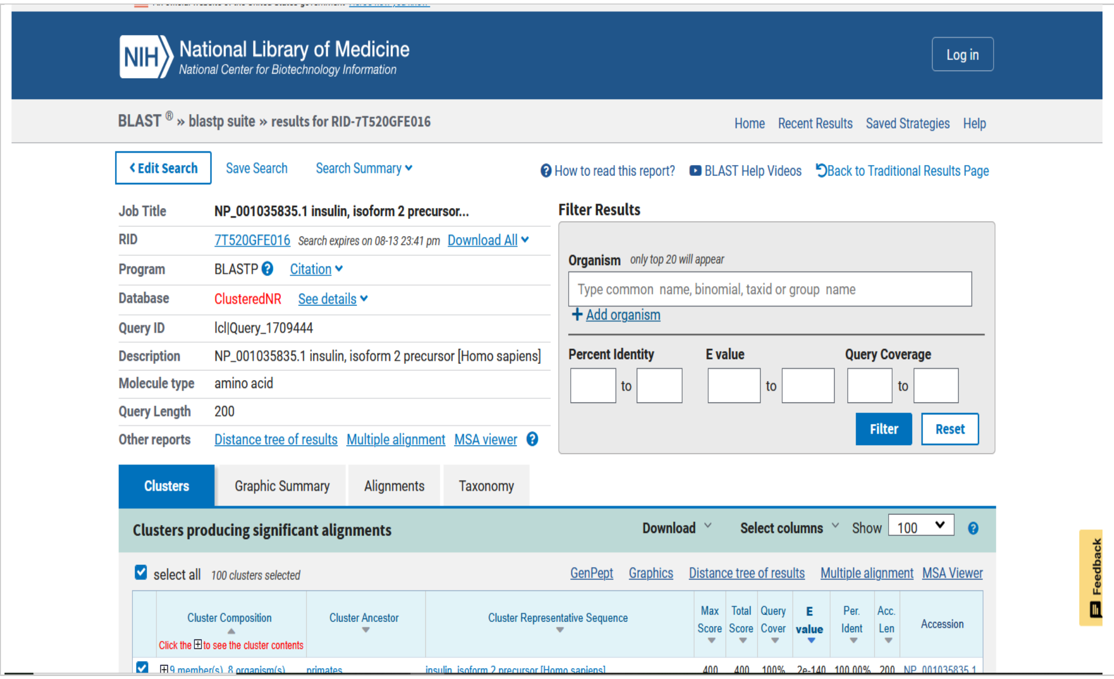
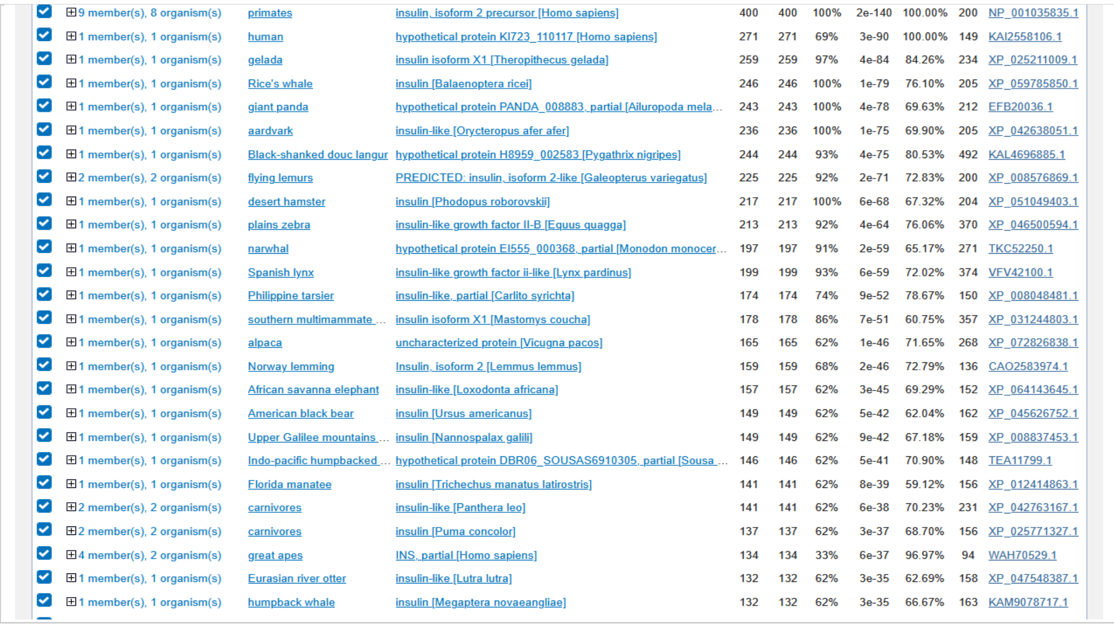


Figure 1. BLASTP job summary for the query NP_001035835.1 (insulin, isoform 2 precursor [Homo sapiens]), run against the ClusteredNR database.



⊞9 member(s), 8 organism(s)	primates	insulin, isoform 2 precursor [Homo sapiens]	400	400	100%	2e-140	100.00%	200	NP_001035835.1
⊞1 member(s), 1 organism(s)	human	hypothetical protein KI723_110117 [Homo sapiens]	271	271	69%	3e-90	100.00%	149	KAI2558106.1
⊞1 member(s), 1 organism(s)	gelada	insulin isoform X1 [Theropithecus gelada]	259	259	97%	4e-84	84.26%	234	XP_025211009.1
⊞1 member(s), 1 organism(s)	Rice's whale	insulin [Balaenoptera ricei]	246	246	100%	1e-79	76.10%	205	XP_059785850.1
⊞1 member(s), 1 organism(s)	giant panda	hypothetical protein PANDA_008883, partial [Ailuropoda mela…	243	243	100%	4e-78	69.63%	212	EFB20036.1
⊞1 member(s), 1 organism(s)	aardvark	insulin-like [Orycteropus afer afer]	236	236	100%	1e-75	69.90%	205	XP_042638051.1
⊞1 member(s), 1 organism(s)	Black-shanked douc langur	hypothetical protein H8959_002583 [Pygathrix nigripes]	244	244	93%	4e-75	80.53%	492	KAL4696885.1
⊞2 member(s), 2 organism(s)	flying lemurs	PREDICTED: insulin, isoform 2-like [Galeopterus variegatus]	225	225	92%	2e-71	72.83%	200	XP_008576869.1
⊞1 member(s), 1 organism(s)	desert hamster	insulin [Phodopus roborovskii]	217	217	100%	6e-68	67.32%	204	XP_051049403.1
⊞1 member(s), 1 organism(s)	plains zebra	insulin-like growth factor II-B [Equus quagga]	213	213	92%	4e-64	76.06%	370	XP_046500594.1
⊞1 member(s), 1 organism(s)	narwhal	hypothetical protein EI555_000368, partial [Monodon monocer…	197	197	91%	2e-59	65.17%	271	TKC52250.1
⊞1 member(s), 1 organism(s)	Spanish lynx	insulin-like growth factor ii-like [Lynx pardinus]	199	199	93%	6e-59	72.02%	374	VFV42100.1
⊞1 member(s), 1 organism(s)	Philippine tarsier	insulin-like, partial [Carlito syrichta]	174	174	74%	9e-52	78.67%	150	XP_008048481.1
⊞1 member(s), 1 organism(s)	southern multimammate …	insulin isoform X1 [Mastomys coucha]	178	178	86%	7e-51	60.75%	357	XP_031244803.1
⊞1 member(s), 1 organism(s)	alpaca	uncharacterized protein [Vicugna pacos]	165	165	62%	1e-46	71.65%	268	XP_072826838.1
⊞1 member(s), 1 organism(s)	Norway lemming	Insulin, isoform 2 [Lemmus lemmus]	159	159	68%	2e-46	72.79%	136	CAO2583974.1
⊞1 member(s), 1 organism(s)	African savanna elephant	insulin-like [Loxodonta africana]	157	157	62%	3e-45	69.29%	152	XP_064143645.1
⊞1 member(s), 1 organism(s)	American black bear	insulin [Ursus americanus]	149	149	62%	5e-42	62.04%	162	XP_045626752.1
⊞1 member(s), 1 organism(s)	Upper Galilee mountains …	insulin [Nannospalax galili]	149	149	62%	9e-42	67.18%	159	XP_008837453.1
⊞1 member(s), 1 organism(s)	Indo-pacific humpbacked …	hypothetical protein DBR06_SOUSAS6910305, partial [Sousa …	146	146	62%	5e-41	70.90%	148	TEA11799.1
⊞1 member(s), 1 organism(s)	Florida manatee	insulin [Trichechus manatus latirostris]	141	141	62%	8e-39	59.12%	156	XP_012414863.1
⊞2 member(s), 2 organism(s)	carnivores	insulin-like [Panthera leo]	141	141	62%	6e-38	70.23%	231	XP_042763167.1
⊞2 member(s), 2 organism(s)	carnivores	insulin [Puma concolor]	137	137	62%	3e-37	68.70%	156	XP_025771327.1
⊞4 member(s), 2 organism(s)	great apes	INS, partial [Homo sapiens]	134	134	33%	6e-37	96.97%	94	WAH70529.1
⊞1 member(s), 1 organism(s)	Eurasian river otter	insulin-like [Lutra lutra]	132	132	62%	3e-35	62.69%	158	XP_047548387.1
⊞1 member(s), 1 organism(s)	humpback whale	insulin [Megaptera novaeangliae]	132	132	62%	3e-35	66.67%	163	KAM9078717.1

Figure 2. Ranked list of clusters producing significant alignments, showing organism, cluster ancestor, score, E-value, and percent identity for each hit.

Table 1. Representative BLAST hits for the human insulin query, spanning close to distant homologs.

Organism	Identity (%)	E-value	Relationship
Homo sapiens (human)	100.00%	2e-140	Query sequence itself (self-match)
Gelada (Theropithecus gelada)	84.26%	4e-84	Old World primate — close relative
Rice's whale (Balaenoptera ricei)	76.10%	1e-79	Placental mammal — moderately close
Aardvark (Orycteropus afer)	69.90%	1e-75	Placental mammal — moderate similarity
Giant panda (Ailuropoda melanoleuca)	69.63%	4e-78	Placental mammal — moderate similarity
Chicken (Gallus gallus)	~30%	9e-22	Distant vertebrate — weak similarity
Klebsiella pneumoniae (bacteria)	34%	7e-21	Non-homologous domain / distant hit

4. Interpretation

The pattern of results aligns closely with what would be expected from evolutionary theory. The query sequence matched itself in the human record with 100% identity, confirming the search was performed correctly. Beyond this, identity percentages decrease in a manner that tracks evolutionary distance from humans:

- Primates such as the gelada (84.26% identity) and other Old World monkeys retained the highest similarity, consistent with a relatively recent common ancestor with humans.
- Other placental mammals — including whales, pandas, aardvarks, and rodents — showed moderate identity, generally in the 60–80% range, reflecting a more distant but still substantial shared ancestry.
- Non-mammalian vertebrates such as chickens showed markedly lower identity (around 30%), and the single bacterial hit (Klebsiella pneumoniae, an insulin-like growth factor domain) showed weak similarity (34%) with a comparatively high E-value, indicating a much more distant or only partially homologous relationship.

All of the strong hits (E-values far below the conventional significance threshold of 0.05, several as low as 2e-140) indicate that these matches are highly unlikely to have occurred by chance and represent genuine biological homology rather than coincidental sequence similarity.

Taken together, these results illustrate that insulin is a highly conserved protein, particularly among mammals. This conservation is expected biologically: insulin performs an essential, tightly regulated metabolic function, so mutations that disrupt its structure are likely to be selected against over evolutionary time. The gradual decline in identity from primates to distantly related species mirrors the branching pattern of the vertebrate evolutionary tree, making insulin a useful marker for studying molecular evolution.

5. Conclusion

This analysis successfully retrieved the human insulin protein sequence from NCBI and used BLASTP to identify a broad set of homologous sequences across species. The results confirm that insulin is strongly conserved among mammals, with identity decreasing progressively in more distantly related organisms, and only weak similarity remaining in non-vertebrate matches. This exercise demonstrates the practical workflow of sequence retrieval and BLAST-based homology searching, and highlights how sequence alignment data can be used to draw meaningful conclusions about protein function and evolutionary relationships.

References

National Center for Biotechnology Information (NCBI). Protein Database. Available at: https://www.ncbi.nlm.nih.gov/protein/

National Center for Biotechnology Information (NCBI). Basic Local Alignment Search Tool (BLAST). Available at: https://blast.ncbi.nlm.nih.gov/